Project Design Phase-II
Data Flow Diagram & User Stories

Date	2 March 2026
Team ID	MCA_DA2
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

DFD Level 0 — Context Diagram

External Input: POSOCO → [India Electricity Dashboard System] → External Output: Policy Analyst / Researcher

Input Data: Daily state-wise electricity consumption (States, Regions, Latitude, Longitude, Dates, Usage in MU)

Output Data: Interactive charts, regional comparisons, lockdown analysis, recovery metrics, story narrative

DFD Level 1 — Process Breakdown

Process	Input	What Happens	Output
P1: Data Ingestion	Consumption.csv	Python validates nulls, data types, date range. Loads to MySQL.	MySQL table: electricity_consumption (16,600 rows)
P2: SQL Aggregation	MySQL raw table	GROUP BY month/region/state. CASE WHEN for lockdown phase. SUM(Usage) for totals.	Monthly totals, regional summaries, phase groups, Top N results
P3: Tableau Connect	MySQL database	Create live/extract connection. Build calculated fields: Month, Year, Quarter, Phase.	Tableau data source (.tds)
P4: Build Charts	Tableau data source	Create 12+ worksheet views: map, line, bar, pie, scatter.	12+ Tableau worksheets
P5: Dashboards	Worksheets	Arrange into 3 dashboards with filters: State, Region, Year, Phase, Top N.	3 Tableau Dashboards
P6: Story	Dashboards + sheets	Build 8-scene story with captions per scene.	1 Tableau Story
P7: Publish	Tableau workbook	Upload to Tableau Public. Get embed URLs.	Tableau Public hosted link

User Stories

Story No.	User Story	Priority	Sprint	Assigned To
USN-1	As a data engineer, I can load Consumption.csv into MySQL and verify 16,600 rows with correct data types.	High	Sprint 1	Divyansh
USN-2	As an analyst, I can run Python validation to confirm 0 nulls, 33 states, 5 regions, correct date range.	High	Sprint 1	Faisal
USN-3	As a BI developer, I can connect Tableau to MySQL and create calculated fields: Month, Year, Quarter, Lockdown Phase.	High	Sprint 2	Faisal
USN-4	As a user, I can view 2019 and 2020 State Consumption bar charts for all 33 states.	High	Sprint 2	Ishita
USN-5	As a user, I can view Total Consumption and Usage by Region charts.	High	Sprint 3	Ishita/Kali
USN-6	As a user, I can use a Top N/Bottom N slider to rank states dynamically.	High	Sprint 3	Kali
USN-7	As a user, I can see a dual-axis 2019 vs 2020 monthly line chart showing the June 2020 lockdown collapse.	High	Sprint 3	Faisal
USN-8	As a user, I can view a Before/After Lockdown bar chart for the top 10 states.	High	Sprint 4	Faisal
USN-9	As a user, I can view Region-wise, Quarter, Metro City, and Year usage charts.	Medium	Sprint 4	Ishita
USN-10	As a user, I can interact with 3 Tableau dashboards with State, Region, Year, and Phase filters.	High	Sprint 5	Faisal
USN-11	As a user, I can navigate a 5-scene Tableau Story covering trend, regional, lockdown, rankings, recovery.	High	Sprint 5	Faisal
USN-12	As a developer, I can publish the workbook to Tableau Public and embed it in the Flask web app.	High	Sprint 6	Divyansh